

Basic Mechanical Engineering (ME2001)

Date: April 3rd 2024

Course Instructor(s)

1. Dr. Kashif Saeed
2. Mohsin Yousuf (CM)

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 40
Total Questions: 2

Roll No _____

Section _____

Student Signature _____

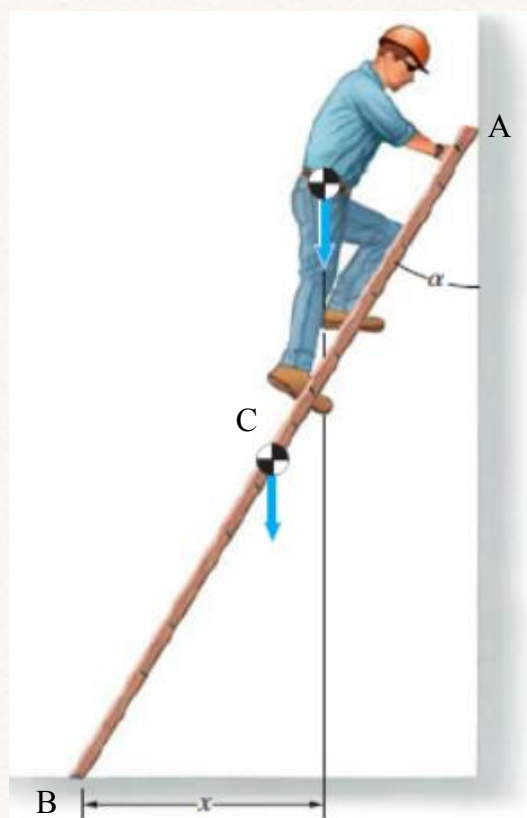
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1. Attempt all the questions, programmable calculators not allowed.
2. Attempt all parts of the same question together.
3. Show all the steps with the help of diagrams and answers with proper units.

CLO # 02: Analyze static equilibrium analysis of a rigid body by applying Newton Laws of Motion and concept of dry friction.

Q1: The ladder and the person weigh 15 kg and 80 kg, respectively. The center of mass of the 3.65 m ladder is at its midpoint (point C) as shown in Fig. 1. The angle $\alpha = 30^\circ$. Assume that the wall exerts a negligible friction force on the ladder at point A.

- (a) If $x = 1.2$ m, **determine** the magnitude of the friction force f_B exerted on the ladder by the floor?
- (b) The person wants to climb to the top the ladder without slipping, **figure out** minimum coefficient of static friction between the ladder and the floor.



[20 marks]

Figure 1. The person on the ladder

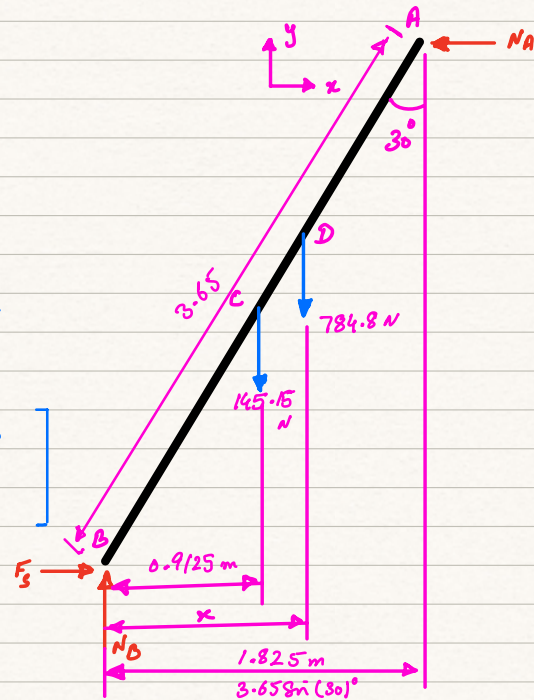
a) $N_B = 931.95 \text{ N}$

Taking moment at A.

$$\begin{bmatrix} -3.65 \sin 30^\circ \\ -3.65 \cos 30^\circ \\ 0 \end{bmatrix} \times \begin{bmatrix} F_s \\ 931.95 \\ 0 \end{bmatrix} + \begin{bmatrix} -0.625 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ -784.8 \\ 0 \end{bmatrix} +$$

$$\begin{bmatrix} -0.9125 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ -145.15 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$F_s = 340.0 \text{ N}$$



b)

$$\begin{bmatrix} -3.65 \sin 30^\circ \\ -3.65 \cos 30^\circ \\ 0 \end{bmatrix} \times \begin{bmatrix} \mu \cdot 931.95 \\ 931.95 \\ 0 \end{bmatrix} + \begin{bmatrix} -1.825 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 145.15 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mu = 0.532$$

CLO # 05: Evaluate the internal forces in the members of a loaded truss and assess safety of the structure.

- Q2:** The structure shown in Fig. 2 is being supported at points *A* and *G*. Use [20 marks]
method of sections to **evaluate** the axial (internal) forces in members *AB*, *BC*
and *CE*. Take $F = 1000$ N.

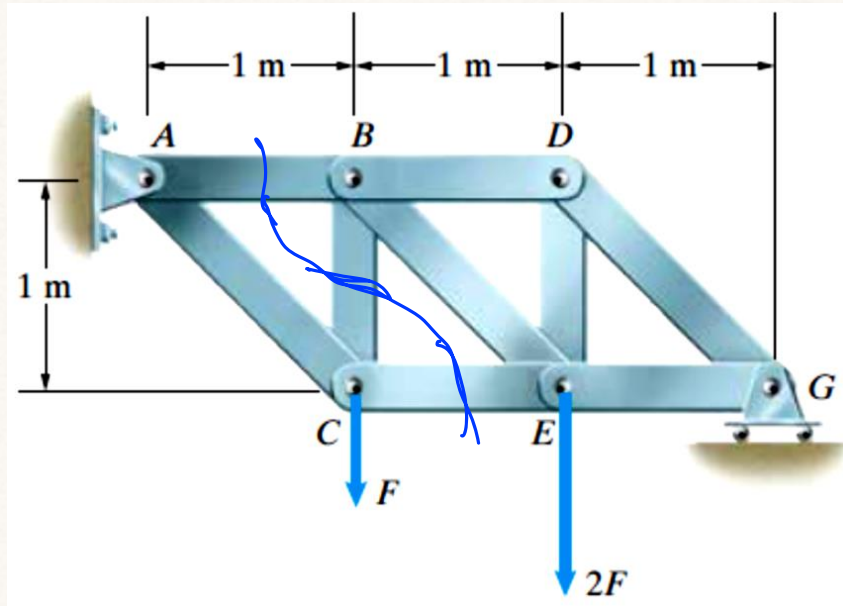
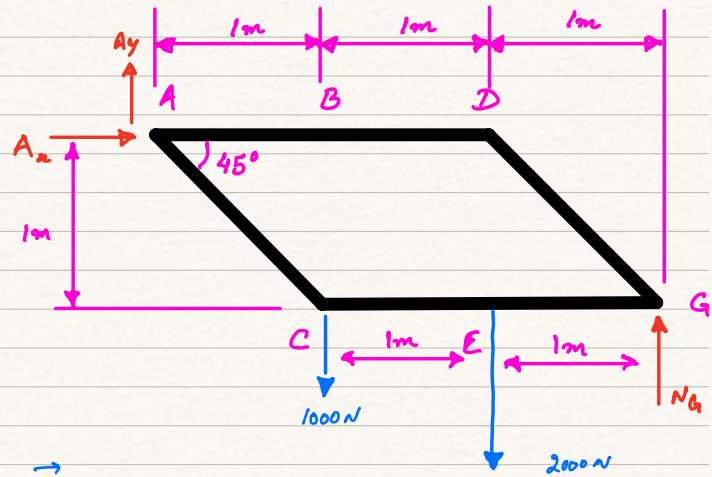


Figure 2. A Truss supported by pin at *A* and roller at *G*



$$\sum \vec{F} = 0$$

$$\begin{bmatrix} A_x \\ A_y \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1000 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -2000 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ N_G \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A_x = 0$$

$$A_y - 1000 - 2000 + N_G = 0$$

$$\sum M_A = 0$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ -1000 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ -2000 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ N_G \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-1000 - 4000 + 3 N_G = 0$$

$$N_G = \frac{5000}{3} = 1666.667 \text{ N}$$

$$\Rightarrow A_y = 1333.333 \text{ N}$$